Aryan Goel

Los Angeles, California | aryangoe@usc.edu | (425) 931-0199 | LinkedIn | GitHub | Portfolio

EDUCATION

University of Southern California — Los Angeles, CA
Master of Science, Computer Science (Data Science Specialization) — *Aug. 2025 – May 2027 (Expected)*
***Relevant Coursework**: Machine Learning for Data Science, Applied Statistical Inference, Deep Learning, Algorithms, Database Systems (indexing, storage engines, query optimization), Foundations of Data Management (Hadoop, Spark, distributed SQL), NLP*

Manipal University Jaipur — Jaipur, India
Bachelor of Technology, Computer & Communication Engineering — GPA: 9.15/10 — *July 2021 – May 2025*

TECHNICAL SKILLS

Languages: Python, SQL, Core Java
Backend & APIs: FastAPI, Supabase, REST API integration, Rate Limiting & Pagination, Caching, Flask, Scapy (packet parsing), Google ADK, Gemini API
Data Systems: Kafka, ClickHouse, TimescaleDB, PostgreSQL, MySQL, DuckDB, Snowflake, PySpark, Dagster, Airflow, dbt, Idempotent Loads
Algorithms: Minimax, Alpha-Beta Pruning, Heuristic Search, Data Structures (coursework: Algorithms)
Testing & DevOps: pytest, Mutation Testing, Data-Contract Tests, GitHub Actions CI, Docker
Tools: Git, VS Code

EXPERIENCE

Indian Institute of Technology (IIT) Roorkee — Roorkee, India
Software Developer — *June 2023 – July 2023*

- Managed and optimized geospatial data layers in **GeoServer** for a production WebGIS application; performed data classification with QGIS
- Redefined the rendering pipeline and resolved performance bugs to reduce load time, while shipping new data-classification features

Indian Institute of Remote Sensing (ISRO) — Dehradun, India
Research Intern — *Feb. 2025 – May 2025*

- Engineered a cascaded CNN architecture (CED-Net) for crop/weed segmentation on RGB drone imagery, outperforming a baseline U-Net through residual blocks and attention mechanisms
- Built the full annotation-to-evaluation pipeline (LabelMe → pixel masks → training/eval on **IoU and Dice score**), benchmarking multiple architecture variants

Indian Institute of Technology (IIT) Roorkee — Roorkee, India
Research Summer Engineer — *May 2024 – July 2024*

- Devised a Physics-Informed Gaussian Process Regression model to forecast flood events, embedding hydrological constraints directly into the model for improved reliability over a standard GPR baseline
- Executed uncertainty quantification via Monte Carlo and Multi-Level Monte Carlo simulations for confidence-bounded risk predictions

PROJECTS

ChainSignal: Crypto Market & On-Chain Anomaly Detection | *Python, PySpark, ClickHouse, Dagster, dbt*

- Built a batch pipeline loading 1,000 coins × 365 days of CoinGecko data and 1.64M Etherscan ERC-20 transfers into ClickHouse, paginating around API limits and verifying loads against raw files (row counts, keys, 256-bit checksums).
- Orchestrated daily incremental and weekly reconcile jobs in Dagster with idempotent partition swaps, a dbt cleaning layer with 28 data-contract tests, 69 pytest tests, and GitHub Actions CI; cut transfer storage 173 → 96 MB via measured ClickHouse codec choices.
- Built no-lookahead PySpark feature jobs and compared three anomaly detectors (robust z-score, Isolation Forest, LOF), evaluated with 887 planted anomalies and a backtest against documented market events.

MarketPulse: Real-Time Trading Anomaly Detector | *Python, Kafka, Spark Structured Streaming, TimescaleDB*

- Built a Kafka + Spark Structured Streaming pipeline over live Finnhub trades (stocks and BTC/ETH), bucketing into 1-minute watermarked event-time bars and persisting every trade to TimescaleDB hypertables.
- Implemented per-ticker stateful detection (log-return, log-volume, trade-rate z-scores against baselines from completed buckets, 0–3 severity, alert cooldown) as a pure, unit-tested function.
- Fixed streaming defects found live (append vs. update output mode, watermark drift, Docker-network Kafka addressing); added a Gemini agent that explains flagged events in plain English.

Packetwright: PCAP Forensics Tool (re-engineered from a course codebase) | *Python, Scapy, DuckDB, Flask*

- Rebuilt the parser as a streaming single-pass engine (Scapy PcapReader) that added ~2 MB of RAM instead of ~865 MB on a 122K-packet capture, with DuckDB storage so multiple investigations coexist.
- Replaced hardcoded thresholds with median/MAD baselines for port-scan and DNS-tunneling detection, with O(1) indexed risk scoring; fixed debug-mode exposure and SYN-ACK double counting.

Little-Go: Adversarial Search Game-Playing AI Agent | *Python, Minimax, Alpha-Beta Pruning*

- Built a Go-playing agent (5×5, USC CSCI-561) with minimax, alpha-beta pruning, and a custom heuristic, winning 60/60 games in the course's grading tournament.
- Wrote a 61-test suite validated by mutation testing (12 planted bugs); measured a 63.5% win rate over the prior version (95% CI, 200–400 games).

BrandSync: Agentic Multi-Modal Ad Generation | *Supabase, FastAPI, Google ADK, Gemini Pro (team project)*

- Designed a **stateless Supabase caching architecture** with per-URL result caching, eliminating redundant compute and API cost entirely on repeat requests
- Built per-asset uploads for cloud deployment and a feedback-routing system that regenerates only the asset an edit concerns, cutting downstream API calls by up to 80%.

ACHIEVEMENTS

- Second Runner-Up, UCLA Glitch × Google DeepMind Hackathon — built BrandSync, then developed it fully after the Google team's positive feedback
- 1st Place, Inter-College Hackathon (Mental Health track) — built a Figma-prototyped app with an integrated chatbot
- Dean's List for Academic Excellence, Manipal University Jaipur (awarded twice)